

BANKING CUSTOMER ANALYTICS & DASHBOARD — PROJECT REPORT

An end-to-end customer data analysis project using Python, PostgreSQL/SQL and Power BI

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1. Project Overview

This project analyzes a retail banking customer dataset covering demographic, account and financial-relationship information for 3,000 clients. The work spans the full analytics pipeline: cleaning and exploring the raw data in Python, storing it in a PostgreSQL database for SQL-based access, and presenting the results through an interactive three-page Power BI dashboard covering customer loans and deposits.

The goal is to turn raw customer records into a clear picture of who the bank's clients are, how their loan and deposit balances vary by branch, occupation, nationality and income level, and to package these insights into a dashboard that business stakeholders can filter and explore on their own.

2. Objectives

- Clean and standardize the raw customer banking dataset and verify data quality (completeness, types, duplicates).
- Engineer an income-band feature to segment clients into Low, Medium and High income groups.
- Explore demographic, relationship and financial variables to uncover patterns and relationships in the client base.
- Persist the cleaned dataset in a relational (PostgreSQL) database to enable SQL-based querying.
- Design an interactive Power BI dashboard that lets stakeholders filter and analyze loans and deposits by branch, gender, occupation, nationality and income band.

3. Tools & Technologies

Category	Tools Used
Data Handling & EDA	Python — pandas, NumPy
Data Visualization (EDA)	Matplotlib, Seaborn
Development Environment	Jupyter Notebook
Database	PostgreSQL
Database Connectivity	SQLAlchemy, psycopg2
Query Language	SQL
Dashboard & Reporting	Power BI Desktop

4. Dataset Overview

Metric	Value
Source file	Banking.csv

Metric	Value
Total records	3,000 clients
Total attributes	25 columns
Missing values	0 — fully complete dataset
Database / table	banking_data.customer_banking_data (PostgreSQL)
Engineered feature	income_band — Low / Medium / High, derived from Estimated Income

Data Dictionary (grouped by category)

Category	Fields
Client profile	Client ID, Name, Age, Location ID, Joined Bank, Banking Contact, Nationality, Occupation, Gender ID
Relationship & risk	Fee Structure, Loyalty Classification, Risk Weighting, Branch ID (BRId), Investment Advisor ID (IAId)
Income & savings	Estimated Income, Income Band, Superannuation Savings
Credit & loans	Amount of Credit Cards, Credit Card Balance, Bank Loans, Business Lending
Deposits & assets	Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Properties Owned

5. Methodology

- **Data ingestion** — Loaded the raw Banking.csv file into a pandas DataFrame and reviewed its structure with `.head()`, `.info()` and `.describe()`.
- **Cleaning & validation** — Standardized all column names to lowercase, underscore-separated form for consistency; confirmed zero missing values across all 3,000 records with `.isnull().sum()`.
- **Feature engineering** — Created the `income_band` column by splitting Estimated Income into three equal-sized groups (Low / Medium / High) using quantile-based binning (`pd.qcut`).
- **Data warehousing** — Wrote the cleaned DataFrame into a PostgreSQL table (`customer_banking_data`, database `banking_data`) via SQLAlchemy, making the dataset queryable with SQL.
- **SQL verification** — Used SQL (see `banking_db.sql`) to query and verify the loaded table, e.g. `SELECT * FROM customer_banking_data LIMIT 20`.
- **Exploratory analysis** — Examined categorical variables with count plots (overall, and split by gender and nationality), numerical variables with histograms, and inter-variable relationships with a correlation heat map.
- **Dashboard design** — Built a 3-page Power BI report (Home, Loan Analysis, Deposit Analysis) with slicers, KPI cards and charts for interactive exploration.

6. Exploratory Data Analysis — Key Insights

6.1 Client Demographics

- Client ages range from 17 to 85 years, averaging around 51 — a broad, evenly-spread customer base rather than one concentrated age group.
- Gender is almost evenly split across the client base (1,512 vs. 1,488 clients in the two recorded categories).
- Nationality is dominated by European clients (1,309 — 43.6%), followed by Asian (754 — 25.1%), American (507 — 16.9%), Australian (254 — 8.5%) and African (176 — 5.9%).

6.2 Relationship, Loyalty & Risk

- Fee Structure skews toward the High tier (1,476 clients — 49.2%), with Mid (962 — 32.1%) and Low (562 — 18.7%) making up the rest.
- Jade is the largest Loyalty Classification (1,331 — 44.4%), ahead of Silver (767), Gold (585) and Platinum (317) — a natural upsell funnel toward higher tiers.
- 64% of clients hold exactly one credit card, 25.5% hold two, and 10.4% hold three.
- Risk Weighting (1–5 scale) is concentrated at the lower end: level 2 is most common (1,222 clients — 40.7%) and level 1 next (836 — 27.9%); only 16% of clients sit at the higher-risk levels 4–5.
- Clients are fairly evenly spread across 0–3 properties owned (roughly 700–780 clients per group).
- Branch 3 (BRId) holds the largest share of clients (1,352 — 45%), well ahead of branches 1, 2 and 4 (493–660 each).

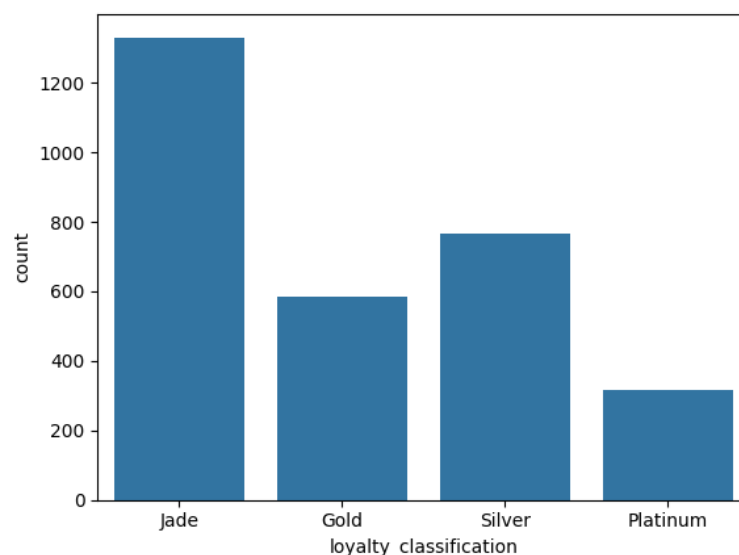


Figure 1 — Client distribution by Loyalty Classification

6.3 Financial Variables & Correlations

- All nine financial variables — income, savings, credit card balance, loans, deposits, checking, saving, foreign-currency and business-lending balances — are right-skewed: most clients cluster at lower values with a long tail of high-value clients.

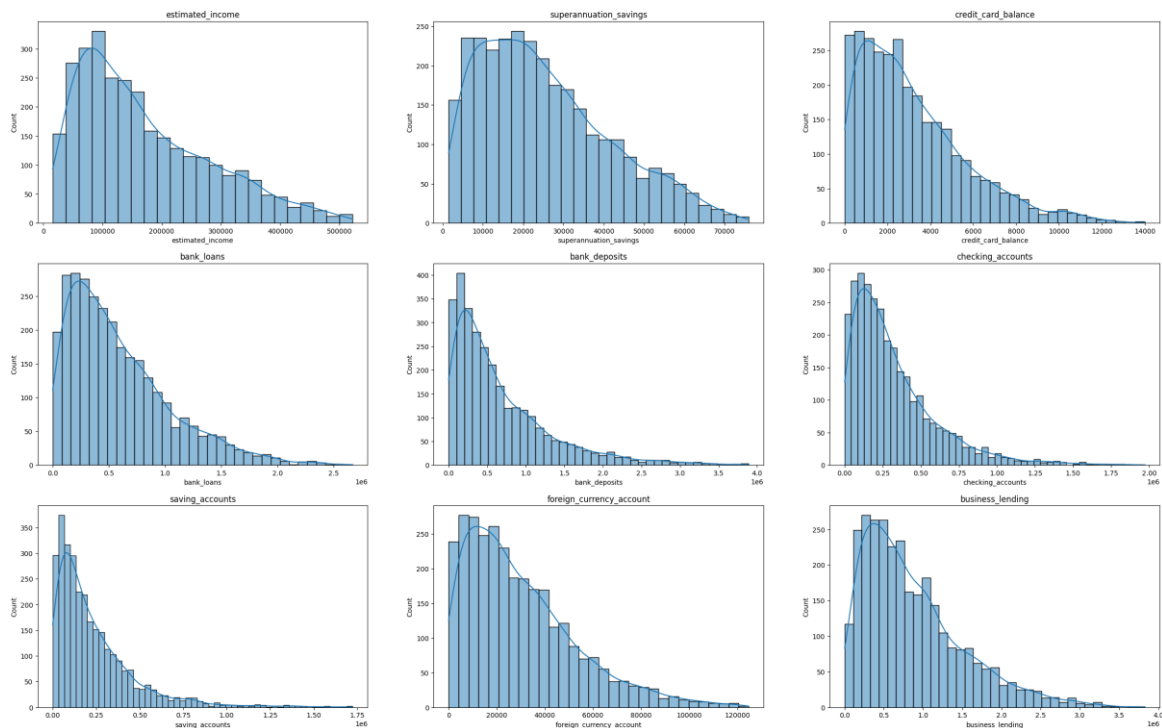


Figure 2 — Distribution of numerical financial variables

- Bank Deposits correlates strongly with Checking Accounts (0.84) and Saving Accounts (0.75), which makes sense since both feed into a client's total deposit balance.
- Every other pair of financial variables shows only mild-to-moderate correlation (0.17–0.44) — each metric captures a fairly distinct part of a client's financial profile, which is useful for segmentation rather than redundant.



Figure 3 — Correlation matrix of financial variables

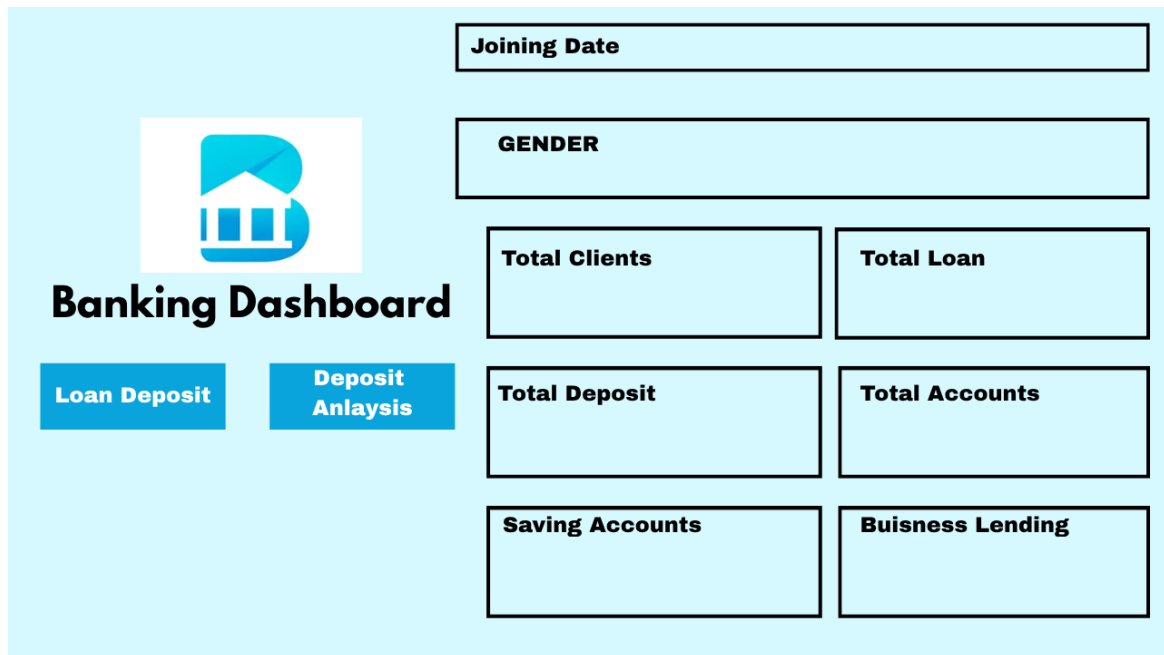
7. Power BI Dashboard

The dashboard is a 3-page interactive Power BI report styled with a consistent light, banking-appropriate theme. Every page shares a top navigation bar, letting users move freely between Home, Loan Analysis and Deposit Analysis.

7.1 Home

Landing page for the report, giving a snapshot of the overall client base before drilling into loans or deposits.

- **Filters (slicers):** Gender, Year of Joining
- **KPI cards:** Total Clients, Total Loan, Total Deposit, Total Accounts, Saving Accounts, Business Lending
- **Charts:** Two navigation buttons routing to the Loan Analysis and Deposit Analysis pages

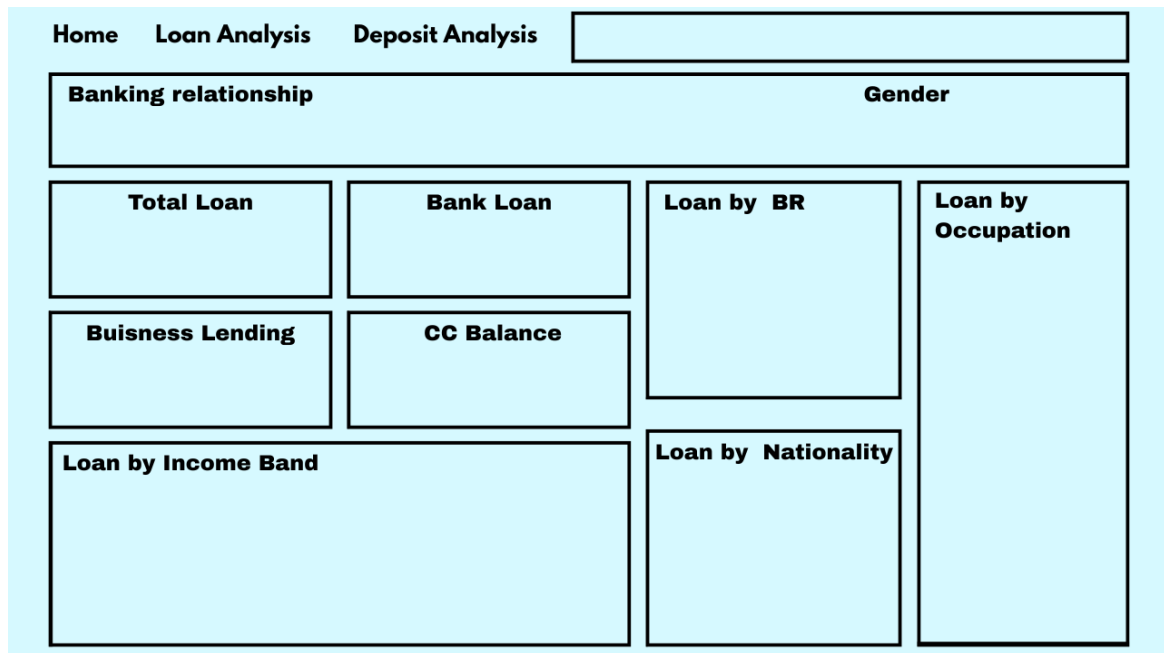


Home page layout

7.2 Loan Analysis

Focused view of the bank's lending exposure, breaking loan values down across the client base.

- **Filters (slicers):** Year of Joining, Gender, Branch (BRId)
- **KPI cards:** Total Loan, Bank Loans, Business Lending, Credit Card Balance
- **Charts:** Loan by Branch, Loan by Occupation and Loan by Nationality (column charts); Loan by Income Band (donut chart)

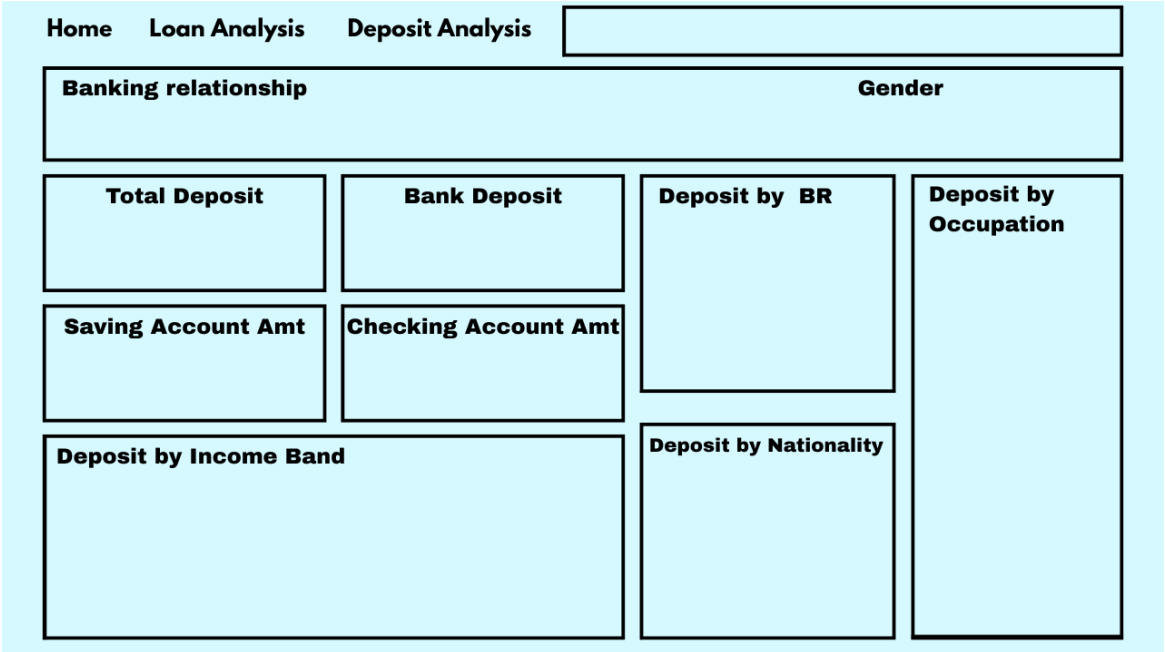


Loan Analysis page layout

7.3 Deposit Analysis

Mirrors the Loan Analysis page for the bank's deposit base, using the same filters for consistency.

- **Filters (slicers):** Gender, Year of Joining, Branch (BRId)
- **KPI cards:** Total Deposit, Bank Deposit, Saving Account Amount, Checking Account Amount
- **Charts:** Deposit by Branch, Deposit by Occupation and Deposit by Nationality (column charts); Deposit by Income Band (donut chart)



Deposit Analysis page layout

8. Key Findings & Business Implications

- Branch 3 carries a disproportionate share of the client base (45%) — worth investigating whether this reflects a larger regional market or an over-reliance on a single branch.
- Nearly half of clients (44.4%) sit in the entry-level Jade loyalty tier, pointing to a sizeable pool of candidates for upgrade campaigns toward Silver, Gold and Platinum.
- The strong link between Bank Deposits, Checking and Saving Accounts confirms these products move together, supporting bundled deposit offers rather than treating them as independent metrics.
- Right-skewed financial variables show a small but valuable segment of high-balance, high-lending clients that stands out clearly in the dashboard and merits closer relationship management.
- With risk weighting concentrated in the low-to-moderate range (68.6% at levels 1–2), the portfolio skews toward lower-risk clients overall, while the smaller high-risk segment (levels 4–5) can be monitored separately.

9. Conclusion

This project demonstrates a complete, practical analytics workflow: cleaning and exploring raw banking data in Python, storing it in a relational database for SQL access, and delivering the results through an interactive Power BI dashboard. Together, the notebook, database and dashboard give a consistent, end-to-end view of the client base — from individual client attributes to branch-level loan and deposit trends — that supports customer segmentation, relationship management and risk-monitoring decisions.

10. Future Scope

- Build a predictive model (e.g., logistic regression or a tree-based classifier) to flag clients at higher risk of default or churn.
- Automate the CSV-to-PostgreSQL load with a scheduled pipeline so the dashboard refreshes on a regular cadence.
- Add drill-through pages from each chart to a client-level detail view.
- Apply row-level security in Power BI so branch managers see only their own branch's data.
- Extend the analysis with clustering (e.g., k-means) on financial variables for a data-driven customer segmentation model, beyond the rule-based income band.